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Проектування розподілених систем

Лабораторна робота 3
«Мікросервіси з використанням Hazelcast Distributed Map»

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Lab 3 - Мікросервіси з використанням Hazelcast Distributed Map

Базова функціональність системи (5 балів)

Клієнт взаємодіє з **facade-service** через HTTP POST та GET запити. У якості клієнта може бути *curl*, *Postman*, браузер у Dev mode та інше

- Разом з запуском **logging-service** запускається новий екземпляр Hazelcast (або клієнт Hazelcast підключається до своєї ноди кластеру Hazelcast)

```
@app.route('/messages', methods=['POST'])
def handle_post():
    """Process POST requests to store messages."""
    logger.info(f"Received POST: {request.form}")
    text = request.form.get('txt')
    if not text:
        logger.warning("Missing txt field")
        return jsonify({"error": "Text required"}), 400

    msg_id = str(uuid.uuid4())
    payload = {"id": msg_id, "txt": text}
    logging_services = retrieve_service_urls("logging")
    if not logging_services:
        logger.error("No logging services available")
        return jsonify({"error": "Logging services unavailable"}), 503

    success, error = post_with_retries(logging_services, payload, msg_id)
    if not success:
        logger.error(f"Failed to process: {error}")
        return jsonify({"error": error}), 500

    logger.info(f"Stored message id={msg_id}")
    return jsonify({"result": "ok", "id": msg_id}), 200
```

- logging-service** використовує Hazelcast Distributed Map для збереження повідомлень замість локальної хеш-таблиці

```
@app.route('/logging', methods=['POST', 'GET'])
def process_logging():
    """Handle logging requests."""
    logger.info(f"[{instance}] Received {request.method} request")
    if request.method == 'POST':
        msg_id = request.form.get('id')
        text = request.form.get('txt')
        if not (msg_id and text):
            logger.warning(f"[{instance}] Missing id or txt")
            return jsonify({"error": "Id and txt required"}), 400

        if message_map.contains_key(msg_id):
```

```

        logger.info(f"[{instance}] Duplicate id={msg_id} detected")
        return jsonify({"message": "Duplicate entry"}), 200

    try:
        message_map.put(msg_id, f"{text} [via {instance}]")
        logger.info(f"[{instance}] Stored id={msg_id}, txt={text}")
        return jsonify({"message": "Stored", "id": msg_id}), 201
    except Exception as e:
        logger.error(f"[{instance}] Failed to store id={msg_id}: {e}")
        return jsonify({"error": str(e)}), 500

```

- **facade-service**, з яким взаємодіє клієнт, випадковим чином обирає екземпляр **logging-service** під час відправки POST/GET (GRPC) запитів якщо обраний екземпляр **logging-service** не доступний, то обирається наступний і т.д.

```

idx = 0
while idx < len(logging_services):
    service = logging_services[idx]
    logger.info(f"Fetching from {service}")
    try:
        response = requests.get(service, timeout=2.5)
        data = response.json()
        results.append({
            "service": service,
            "instance": data.get("instance"),
            "messages": data.get("messages")
        })
        logger.info(f"Retrieved from {service}: {data.get('messages')}")
        break
    except requests.RequestException as e:
        logger.error(f"Failed to fetch from {service}: {e}")
        results.append({"service": service, "error": str(e)})
        idx += 1

```

Додатковий функціонал системи (+5 балів)

У зв'язку з тим, що тепер може бути декілька запущених екземплярів **logging-service**, то **facade-service** має знати про їх IP-адреси, для доступу до них. Перелік IP-адрес, може міститись у коді самого **facade-service**, чи передаватись у нього при старті, проте такий підхід є не гнучким, у разі динамічного призначення чм зміни цих адрес.

Тому, пропонується винести інформацію про IP-адреси інших мікросервісів у окремий реєстр, за який буде відповідати **config-server** (пізніше він буде замінений на *Service registry and discovery*).

Тепер, перед зверненням до **logging-service** та **messages-service**, **facade-service** має робити запит до **config-server**, де по імені сервіса йому повертається перелік всіх IP-адрес екземплярів даного сервіса.

Перелік IP-адрес мікросервісів на **config-server**, може братись з конфігураційного файлу чи передаватись з командного рядка під час його запуску.

Ініціалізація адрес:

```
logging_urls = os.environ.get("LOGGING_URLS",  
"logging1:50051,logging2:50052,logging3:50053").split(',')  
messages_urls = os.environ.get("MESSAGES_URLS", "messages:8882").split(',')  
  
logging_urls = [f"http://{url}/logging" for url in logging_urls]  
messages_urls = [f"http://{url}/messages" for url in messages_urls]
```

Ендпоінт для повернення адрес:

```
@app.route('/services/<service_name>', methods=['GET'])  
  
def retrieve_service_urls(service_name):  
    """Return URLs for the requested service."""  
    logger.info(f"Query for {service_name} URLs")  
    if service_name == "logging":  
        return jsonify({"urls": logging_urls}), 200  
    elif service_name == "messages":  
        return jsonify({"urls": messages_urls}), 200  
    logger.warning(f"Service {service_name} not found")  
    return jsonify({"error": f"Service {service_name} not found"}), 404
```

У docker-compose.yml для config-server визначено змінні середовища:

```
environment:  
  - CFG_PORT=8883  
  - LOGGING_URLS=logging1:50051,logging2:50052,logging3:50053  
  - MESSAGES_URLS=messages:8882
```

facade-service робить запит до config-server перед зверненням до logging-service та messages-service

```
def retrieve_service_urls(service):  
    """Fetch service URLs from config-server."""  
    full_url = f"{CONFIG_URL}/{service}"  
    logger.info(f"Requesting URLs from {full_url}")  
    try:  
        response = requests.get(full_url, timeout=2.5)  
        response.raise_for_status()  
        urls = response.json().get("urls", [])  
        if not urls:  
            logger.error(f"No URLs returned for {service}")  
            logger.info(f"Retrieved URLs for {service}: {urls}")  
            return urls  
    except requests.RequestException as e:  
        logger.error(f"Failed to fetch URLs for {service} from {full_url}: {e}")  
        return []
```

У config-server.py адреси беруться із змінних середовища:

```
logging_urls = os.environ.get("LOGGING_URLS",  
"logging1:50051,logging2:50052,logging3:50053").split(',')
```

```
messages_urls = os.environ.get("MESSAGES_URLS", "messages:8882").split(',')
```

Завдання

- Запустити три екземпляра **logging-service** (локально їх можна запустити на різних портах), відповідно мають запуститись також три екземпляра Hazelcast

☐	Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	task3	-	-	-	3.86%	3 seconds ago	
☐	hazelcast1	cdf6e6856046	hazelcast/hazelcast:5.3	5701:5701 ↗	1.13%	10 seconds ago	
☐	hazelcast2	8e869c16f6c8	hazelcast/hazelcast:5.3	5702:5701 ↗	1.26%	10 seconds ago	
☐	hazelcast3	cc297596fe21	hazelcast/hazelcast:5.3	5703:5701 ↗	1.47%	10 seconds ago	
☐	config-server	4fa489dbf963	task3-config-server	8883:8883 ↗	0%	10 seconds ago	
☐	logging2	dcc45ef9f0fb	task3-logging2	50052:50052 ↗	0%	4 seconds ago	
☐	logging3	3318ebfb4f64	task3-logging3	50053:50053 ↗	0%	4 seconds ago	
☐	logging1	4dd92eabcb19	task3-logging1	50051:50051 ↗	0%	4 seconds ago	
Showing 10 items							
☐	messages	5c2020ba967a	task3-messages	8882:8882 ↗	0.92%	19 seconds ago	
☐	facade	9372a235bf0b ↗	task3-facade	8880:8880 ↗	0.21%	18 seconds ago	
Showing 10 items							

- Через HTTP POST записати 10 повідомлень **msg1-msg10** через **facade-service**

```
PS D:\dist-sys\Task3> curl.exe -X POST http://localhost:8880/messages -F "txt=message1"
{
  "id": "0b2c95e5-c4d9-4d6a-8415-a6d3ec3dcac1",
  "result": "ok"
}
PS D:\dist-sys\Task3> curl.exe -X POST http://localhost:8880/messages -F "txt=message2"
{
  "id": "0bf5e357-a203-439c-bd00-2bea746c0949",
  "result": "ok"
}
PS D:\dist-sys\Task3> curl.exe -X POST http://localhost:8880/messages -F "txt=message3"
{
  "id": "1959c02d-2464-4d6a-a805-29dcd835cb36",
  "result": "ok"
}
PS D:\dist-sys\Task3> curl.exe -X POST http://localhost:8880/messages -F "txt=message4"
{
  "id": "1183fe9a-9b01-48ff-a57a-9867561386a4",
  "result": "ok"
}
PS D:\dist-sys\Task3> curl.exe -X POST http://localhost:8880/messages -F "txt=message5"
{
  "id": "e09fdc53-clac-4b56-a42a-cfd954c87c72",
  "result": "ok"
}
PS D:\dist-sys\Task3> curl.exe -X POST http://localhost:8880/messages -F "txt=message6"
{
  "id": "126362e0-3f9f-4731-a51c-32128f7288c6",
  "result": "ok"
}
PS D:\dist-sys\Task3> curl.exe -X POST http://localhost:8880/messages -F "txt=message7"
{
  "id": "8570d74d-55a6-41df-bf68-e9e6691284d0",
  "result": "ok"
}
PS D:\dist-sys\Task3> curl.exe -X POST http://localhost:8880/messages -F "txt=message8"
{
  "id": "34f281fb-c45d-4c76-ac41-295a0308015c",
  "result": "ok"
}
PS D:\dist-sys\Task3> curl.exe -X POST http://localhost:8880/messages -F "txt=message9"
{
  "id": "a2b237da-d4df-4047-873b-a575306ff7ee",
  "result": "ok"
}
PS D:\dist-sys\Task3> curl.exe -X POST http://localhost:8880/messages -F "txt=message10"
{
  "id": "b3b08728-6a97-4e9d-bc08-0f1cd692a3fa",
  "result": "ok"
}
```

- Показати які повідомлення отримав кожен з екземплярів **logging-service** (це має бути видно у логах сервісу)
Logging1:

```

PS D:\dist-sys\Task3> docker-compose logs logging1
time="2025-06-16T17:11:19+03:00" level=warning msg="D:\dist-sys\Task3\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid pote
ntial confusion"
logging1 2025-06-16 14:09:18,544 [LOG] HazelcastClient 5.5.0 is STARTING
logging1 2025-06-16 14:09:18,544 [LOG] HazelcastClient 5.5.0 is STARTED
logging1 2025-06-16 14:09:18,544 [LOG] Trying to connect to Address(host=hazelcast1, port=5701)
logging1 2025-06-16 14:09:20,589 [LOG] HazelcastClient 5.5.0 is CONNECTED
logging1 2025-06-16 14:09:20,589 [LOG] Authenticated with server Address(host=172.18.0.3, port=5701):c79b5d48-dc91-4f51-bcd3-5fea83c10e80, server version: 5.3.8, local address:
Address(host=172.18.0.8, port=48296)
logging1 2025-06-16 14:09:20,593 [LOG]
logging1
logging1 Members [3] {
logging1 | Member [172.18.0.5]:5701 - 84ecac0d-d35b-4b57-a3ca-0f14e8cabf72
logging1 | Member [172.18.0.2]:5701 - 08ca836f-2117-4bdd-8758-4b62c5a5e694
logging1 | Member [172.18.0.3]:5701 - c79b5d48-dc91-4f51-bcd3-5fea83c10e80
logging1 | }
logging1
logging1 2025-06-16 14:09:20,598 [LOG] Authenticated with server Address(host=172.18.0.5, port=5701):84ecac0d-d35b-4b57-a3ca-0f14e8cabf72, server version: 5.3.8, local address:
Address(host=172.18.0.8, port=46328)
logging1 2025-06-16 14:09:20,601 [LOG] Authenticated with server Address(host=172.18.0.2, port=5701):08ca836f-2117-4bdd-8758-4b62c5a5e694, server version: 5.3.8, local address:
Address(host=172.18.0.8, port=32439)
logging1 2025-06-16 14:09:20,603 [LOG] Client started
logging1 2025-06-16 14:09:20,606 [LOG] [b903ac56b109] Connected to Hazelcast cluster
logging1 2025-06-16 14:09:20,607 [LOG] [b903ac56b109] Starting logging-service on port 50051
logging1 * Serving Flask app 'logging-service'
logging1 * Debug mode: off
logging1 2025-06-16 14:09:20,611 [LOG] WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
logging1 * Running on all addresses (0.0.0.0)
logging1 * Running on http://172.18.0.1:50051
logging1 * Running on http://172.18.0.8:50051
logging1 2025-06-16 14:09:20,611 [LOG] Press CTRL+C to quit
logging1 2025-06-16 14:10:09,088 [LOG] [b903ac56b109] Received POST request
logging1 2025-06-16 14:10:09,153 [LOG] [b903ac56b109] Stored id=0bf5e357-a203-439c-bd00-2bea746c0949, txt=message2
logging1 2025-06-16 14:10:09,153 [LOG] 172.18.0.10 - - [16/Jun/2025 14:10:09] "POST /logging HTTP/1.1" 201 -
logging1 2025-06-16 14:10:12,554 [LOG] [b903ac56b109] Received POST request
logging1 2025-06-16 14:10:12,559 [LOG] [b903ac56b109] Stored id=1959c02d-2464-4d6a-a805-29dcd835cb36, txt=message3
logging1 2025-06-16 14:10:12,560 [LOG] 172.18.0.10 - - [16/Jun/2025 14:10:12] "POST /logging HTTP/1.1" 201 -
logging1 2025-06-16 14:10:15,452 [LOG] [b903ac56b109] Received POST request
logging1 2025-06-16 14:10:15,455 [LOG] [b903ac56b109] Stored id=1183fe9a-0b01-48ff-a57a-98675613864a, txt=message4
logging1 2025-06-16 14:10:15,455 [LOG] 172.18.0.10 - - [16/Jun/2025 14:10:15] "POST /logging HTTP/1.1" 201 -
logging1 2025-06-16 14:10:24,688 [LOG] [b903ac56b109] Received POST request
logging1 2025-06-16 14:10:24,691 [LOG] [b903ac56b109] Stored id=34f281fb-c45d-4c76-ac41-295a0308015c, txt=message8
logging1 2025-06-16 14:10:24,692 [LOG] 172.18.0.10 - - [16/Jun/2025 14:10:24] "POST /logging HTTP/1.1" 201 -
logging1 2025-06-16 14:10:27,391 [LOG] [b903ac56b109] Received POST request
logging1 2025-06-16 14:10:27,395 [LOG] [b903ac56b109] Stored id=a2b237da-d4df-4047-873b-a575306ff7ee, txt=message9
logging1 2025-06-16 14:10:27,395 [LOG] 172.18.0.10 - - [16/Jun/2025 14:10:27] "POST /logging HTTP/1.1" 201 -
logging1 2025-06-16 14:10:29,918 [LOG] [b903ac56b109] Received POST request
logging1 2025-06-16 14:10:29,921 [LOG] [b903ac56b109] Stored id=b3080728-6a97-4e9d-bc08-0f1cd692a3fa, txt=message10
logging1 2025-06-16 14:10:29,922 [LOG] 172.18.0.10 - - [16/Jun/2025 14:10:29] "POST /logging HTTP/1.1" 201 -
PS D:\dist-sys\Task3>

```

Logging2:

```

PS D:\dist-sys\Task3> docker-compose logs logging2
time="2025-06-16T17:11:44+03:00" level=warning msg="D:\dist-sys\Task3\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid pote
ntial confusion"
logging2 2025-06-16 14:09:18,508 [LOG] HazelcastClient 5.5.0 is STARTING
logging2 2025-06-16 14:09:18,508 [LOG] HazelcastClient 5.5.0 is STARTED
logging2 2025-06-16 14:09:18,509 [LOG] Trying to connect to Address(host=hazelcast2, port=5701)
logging2 2025-06-16 14:09:18,527 [LOG] HazelcastClient 5.5.0 is CONNECTED
logging2 2025-06-16 14:09:18,527 [LOG] Authenticated with server Address(host=172.18.0.5, port=5701):84ecac0d-d35b-4b57-a3ca-0f14e8cabf72, server version: 5.3.8, local address:
Address(host=172.18.0.7, port=36286)
logging2 2025-06-16 14:09:18,534 [LOG]
logging2
logging2 Members [1] {
logging2 | Member [172.18.0.5]:5701 - 84ecac0d-d35b-4b57-a3ca-0f14e8cabf72
logging2 | }
logging2
logging2 2025-06-16 14:09:18,539 [LOG] Client started
logging2 2025-06-16 14:09:18,546 [LOG] [bd5809e31194] Connected to Hazelcast cluster
logging2 2025-06-16 14:09:18,547 [LOG] [bd5809e31194] Starting logging-service on port 50052
logging2 * Serving Flask app 'logging-service'
logging2 * Debug mode: off
logging2 2025-06-16 14:09:18,551 [LOG] WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
logging2 * Running on all addresses (0.0.0.0)
logging2 * Running on http://172.18.0.1:50052
logging2 * Running on http://172.18.0.7:50052
logging2 2025-06-16 14:09:18,551 [LOG] Press CTRL+C to quit
logging2 2025-06-16 14:09:20,115 [LOG]
logging2
logging2 Members [3] {
logging2 | Member [172.18.0.5]:5701 - 84ecac0d-d35b-4b57-a3ca-0f14e8cabf72
logging2 | Member [172.18.0.2]:5701 - 08ca836f-2117-4bdd-8758-4b62c5a5e694
logging2 | Member [172.18.0.3]:5701 - c79b5d48-dc91-4f51-bcd3-5fea83c10e80
logging2 | }
logging2
logging2 2025-06-16 14:09:20,558 [LOG] Authenticated with server Address(host=172.18.0.2, port=5701):08ca836f-2117-4bdd-8758-4b62c5a5e694, server version: 5.3.8, local address:
Address(host=172.18.0.7, port=45904)
logging2 2025-06-16 14:09:20,558 [LOG] Authenticated with server Address(host=172.18.0.3, port=5701):c79b5d48-dc91-4f51-bcd3-5fea83c10e80, server version: 5.3.8, local address:
Address(host=172.18.0.7, port=58308)
logging2 2025-06-16 14:10:19,902 [LOG] [bd5809e31194] Received POST request
logging2 2025-06-16 14:10:19,907 [LOG] [bd5809e31194] Stored id=126362e0-3f9f-4731-a51c-32128f7288c6, txt=message6
logging2 2025-06-16 14:10:19,908 [LOG] 172.18.0.10 - - [16/Jun/2025 14:10:19] "POST /logging HTTP/1.1" 201 -
logging2 2025-06-16 14:10:22,072 [LOG] [bd5809e31194] Received POST request
logging2 2025-06-16 14:10:22,076 [LOG] [bd5809e31194] Stored id=8570d74d-55a6-41df-bf68-e9e6691284d0, txt=message7
logging2 2025-06-16 14:10:22,076 [LOG] 172.18.0.10 - - [16/Jun/2025 14:10:22] "POST /logging HTTP/1.1" 201 -
PS D:\dist-sys\Task3>

```

Logging3:

```

PS D:\dist-sys\Task3> docker-compose logs logging3
time="2025-06-16T17:12:09+03:00" level=warning msg="D:\dist-sys\Task3\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid pote
ntial confusion"
logging3 2025-06-16 14:09:18,544 [LOG] HazelcastClient 5.5.0 is STARTING
logging3 2025-06-16 14:09:18,545 [LOG] HazelcastClient 5.5.0 is STARTED
logging3 2025-06-16 14:09:18,545 [LOG] Trying to connect to Address(host=hazelcast3, port=5701)
logging3 2025-06-16 14:09:20,582 [LOG] HazelcastClient 5.5.0 is CONNECTED
logging3 2025-06-16 14:09:20,582 [LOG] Authenticated with server Address(host=172.18.0.2, port=5701):08ca836f-2117-4bdd-8758-4b62c5a5e694, server version: 5.3.8, local address:
Address(host=172.18.0.6, port=43266)
logging3 2025-06-16 14:09:20,586 [LOG]
logging3
logging3 Members [3] {
logging3 | Member [172.18.0.5]:5701 - 84ecac0d-d35b-4b57-a3ca-0f14e8cabf72
logging3 | Member [172.18.0.2]:5701 - 08ca836f-2117-4bdd-8758-4b62c5a5e694
logging3 | Member [172.18.0.3]:5701 - c79b5d48-dc91-4f51-bcd3-5fea83c10e80
logging3 | }
logging3
logging3 2025-06-16 14:09:20,592 [LOG] Authenticated with server Address(host=172.18.0.5, port=5701):84ecac0d-d35b-4b57-a3ca-0f14e8cabf72, server version: 5.3.8, local address:
Address(host=172.18.0.6, port=56394)
logging3 2025-06-16 14:09:20,594 [LOG] Authenticated with server Address(host=172.18.0.3, port=5701):c79b5d48-dc91-4f51-bcd3-5fea83c10e80, server version: 5.3.8, local address:
Address(host=172.18.0.6, port=46318)
logging3 2025-06-16 14:09:20,598 [LOG] Client started
logging3 2025-06-16 14:09:20,600 [LOG] [5fa37145289e] Connected to Hazelcast cluster
logging3 2025-06-16 14:09:20,601 [LOG] [5fa37145289e] Starting logging-service on port 50053
logging3 * Serving Flask app 'logging-service'
logging3 * Debug mode: off
logging3 2025-06-16 14:09:20,606 [LOG] WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
logging3 * Running on all addresses (0.0.0.0)
logging3 * Running on http://172.18.0.1:50053
logging3 * Running on http://172.18.0.6:50053
logging3 2025-06-16 14:09:20,606 [LOG] Press CTRL+C to quit
logging3 2025-06-16 14:10:03,910 [LOG] [5fa37145289e] Received POST request
logging3 2025-06-16 14:10:04,069 [LOG] [5fa37145289e] Stored id=0b2c95c5-c4d9-4d6a-8415-a6d3ec3dcacl, txt=message1
logging3 2025-06-16 14:10:04,069 [LOG] 172.18.0.10 - - [16/Jun/2025 14:10:04] "POST /logging HTTP/1.1" 201 -
logging3 2025-06-16 14:10:17,679 [LOG] [5fa37145289e] Received POST request
logging3 2025-06-16 14:10:17,684 [LOG] [5fa37145289e] Stored id=e09fd533-clac-4b56-a42a-cfd954c87c72, txt=message5
logging3 2025-06-16 14:10:17,684 [LOG] 172.18.0.10 - - [16/Jun/2025 14:10:17] "POST /logging HTTP/1.1" 201 -
PS D:\dist-sys\Task3>

```

- Через HTTP GET з *facade-service* прочитати повідомлення

```
PS D:\dist-sys\Task3> curl.exe http://localhost:8880/messages
{
  "all_messages": [
    {
      "id": "a2b237da-d4df-4047-873b-a575306ff7ee",
      "text": "message9 [via b903ac56b109]"
    },
    {
      "id": "126362e0-3f9f-4731-a51c-32128f7288c6",
      "text": "message6 [via bd5809e31194]"
    },
    {
      "id": "e09fdc53-clac-4b56-a42a-cfd954c87c72",
      "text": "message5 [via 5fa37145289e]"
    },
    {
      "id": "0b2c95e5-c4d9-4d6a-8415-a6d3ac3dcac1",
      "text": "message1 [via 5fa37145289e]"
    },
    {
      "id": "8570d74d-55a6-41df-bf68-e9e6691284d0",
      "text": "message7 [via bd5809e31194]"
    },
    {
      "id": "0bf5e357-a203-439c-bd00-2bea746c0949",
      "text": "message2 [via b903ac56b109]"
    },
    {
      "id": "34f281fb-c45d-4c76-ac41-295a0308015c",
      "text": "message8 [via b903ac56b109]"
    },
    {
      "id": "b3b08728-6a97-4e9d-bc08-0f1cd692a3fa",
      "text": "message10 [via b903ac56b109]"
    },
    {
      "id": "1959c02d-2464-4d6a-a805-29dcd835cb36",
      "text": "message3 [via b903ac56b109]"
    },
    {
      "id": "1183fe9a-9b01-48ff-a57a-9867561386a4",
      "text": "message4 [via b903ac56b109]"
    }
  ],
  "logging_services": [
    {
      "instance": "5fa37145289e",
      "messages": [
        "message9 [via b903ac56b109]",
        "message6 [via bd5809e31194]",
        "message5 [via 5fa37145289e]",
        "message1 [via 5fa37145289e]",
        "message7 [via bd5809e31194]",
        "message2 [via b903ac56b109]",
        "message8 [via b903ac56b109]",
        "message10 [via b903ac56b109]",
        "message3 [via b903ac56b109]",
        "message4 [via b903ac56b109]"
      ],
      "service": "http://logging3:50053/logging"
    }
  ]
}
```

- Вимкнути один/два екземпляри **logging-service** (разом з ним мають вимикатись й ноди Hazelcast) та перевірити чи зможемо прочитати повідомлення

```
PS D:\dist-sys\Task3> docker-compose stop logging1 logging2 hazelcast1 hazelcast2
time="2025-06-16T17:40:14+03:00" level=warning msg="D:\\dist-sys\\Task3\\docker-compose.yml
[+] Stopping 4/4
✓Container logging1      Stopped
✓Container logging2      Stopped
✓Container hazelcast1     Stopped
✓Container hazelcast2     Stopped
PS D:\dist-sys\Task3> |
```

☐	Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	 hazelcast2	ccacca157fc6	hazelcast/hazelcast:5.3	5702:5701	0%	33 minutes ago	  
☐	 config-server	e52f8c53ae6b	task3-config-server	8883:8883 	1.56%	33 minutes ago	  
☐	 logging3	5fa37145289e	task3-logging3	50053:50053 	0.85%	32 minutes ago	  
☐	 messages	2070ee734e4c	task3-messages	8882:8882 	0.79%	32 minutes ago	  
☐	 hazelcast1	7e9cf9432ce6	hazelcast/hazelcast:5.3	5701:5701	0%	5 minutes ago	  
☐	 logging1	b903ac56b109	task3-logging1	50051:50051	0%	5 minutes ago	  
☐	 logging2	bd5809e31194	task3-logging2	50052:50052	0%	32 minutes ago	  
☐	 facade	e38fa0e73e22	task3-facade	8880:8880 	0.11%	32 minutes ago	  

Showing 10 items

Логи façade:

Відправка повідомлень без двох екземплярів Hazelcast та нодів:


```
Windows PowerShell
facade | 2025-06-16 14:51:12.999 [FCD] Received POST: ImmutableMultiDict({'txt': 'message1'})
facade | 2025-06-16 14:51:13.000 [FCD] Requesting URLs from http://config-server:8883/services/logging
facade | 2025-06-16 14:51:13.012 [FCD] Retrieved URLs for Logging: ['http://logging1:50051/logging', 'http://logging2:50052/logging', 'http://logging3:50053/logging']
facade | 2025-06-16 14:51:13.012 [FCD] Attempting to send message id=8c437f76-07d1-45a8-b38f-46d5b61b97d1
facade | 2025-06-16 14:51:13.012 [FCD] Attempt 1 to http://logging2:50052/logging
facade | 2025-06-16 14:51:16.785 [FCD] Error sending to http://logging2:50052/logging: HTTPConnectionPool(host='logging2', port=50052): Max retries exceeded with url: /logging (Ca
used by NameResolutionError("curllib3.connection.HTTPConnection object at 0x7fca57b57160: Failed to resolve 'logging2' ([Errno -2] Name or service not known)"))
facade | 2025-06-16 14:51:16.785 [FCD] Waiting 0.5s before retry
facade | 2025-06-16 14:51:17.286 [FCD] Attempt 2 to http://logging3:50053/logging
facade | 2025-06-16 14:51:17.296 [FCD] Successfully sent to http://logging3:50053/logging
facade | 2025-06-16 14:51:17.296 [FCD] Stored message id=8c437f76-07d1-45a8-b38f-46d5b61b97d1
facade | 2025-06-16 14:51:17.297 [FCD] 172.18.0.1 - - [16/Jun/2025 14:51:17] "POST /messages HTTP/1.1" 200 -
facade | 2025-06-16 14:51:21.185 [FCD] Received POST: ImmutableMultiDict({'txt': 'message12'})
facade | 2025-06-16 14:51:21.185 [FCD] Requesting URLs from http://config-server:8883/services/logging
facade | 2025-06-16 14:51:21.189 [FCD] Retrieved URLs for Logging: ['http://logging1:50051/logging', 'http://logging2:50052/logging', 'http://logging3:50053/logging']
facade | 2025-06-16 14:51:21.190 [FCD] Attempting to send message id=1eb3b4e2-1344-4253-9dca-b9171f784941
facade | 2025-06-16 14:51:21.190 [FCD] Attempt 1 to http://logging2:50052/logging
facade | 2025-06-16 14:51:24.966 [FCD] Error sending to http://logging2:50052/logging: HTTPConnectionPool(host='logging2', port=50052): Max retries exceeded with url: /logging (Ca
used by NameResolutionError("curllib3.connection.HTTPConnection object at 0x7fca57bb03a0: Failed to resolve 'logging2' ([Errno -2] Name or service not known)"))
facade | 2025-06-16 14:51:24.966 [FCD] Waiting 0.5s before retry
facade | 2025-06-16 14:51:25.467 [FCD] Attempt 2 to http://logging3:50053/logging
facade | 2025-06-16 14:51:25.477 [FCD] Successfully sent to http://logging3:50053/logging
facade | 2025-06-16 14:51:25.477 [FCD] Stored message id=1eb3b4e2-1344-4253-9dca-b9171f784941
facade | 2025-06-16 14:51:25.477 [FCD] 172.18.0.1 - - [16/Jun/2025 14:51:25] "POST /messages HTTP/1.1" 200 -
facade | 2025-06-16 14:51:27.475 [FCD] Received POST: ImmutableMultiDict({'txt': 'message13'})
facade | 2025-06-16 14:51:27.475 [FCD] Requesting URLs from http://config-server:8883/services/logging
facade | 2025-06-16 14:51:27.480 [FCD] Retrieved URLs for Logging: ['http://logging1:50051/logging', 'http://logging2:50052/logging', 'http://logging3:50053/logging']
facade | 2025-06-16 14:51:27.480 [FCD] Attempting to send message id=100581f3-7743-4799-84d0-a78c295c4e2b
facade | 2025-06-16 14:51:27.481 [FCD] Attempt 1 to http://logging2:50052/logging
facade | 2025-06-16 14:51:31.247 [FCD] Error sending to http://logging2:50052/logging: HTTPConnectionPool(host='logging2', port=50052): Max retries exceeded with url: /logging (Ca
used by NameResolutionError("curllib3.connection.HTTPConnection object at 0x7fca57b57190: Failed to resolve 'logging2' ([Errno -2] Name or service not known)"))
facade | 2025-06-16 14:51:31.248 [FCD] Waiting 0.5s before retry
facade | 2025-06-16 14:51:31.749 [FCD] Attempt 2 to http://logging3:50053/logging
facade | 2025-06-16 14:51:31.758 [FCD] Successfully sent to http://logging3:50053/logging
facade | 2025-06-16 14:51:31.758 [FCD] Stored message id=100581f3-7743-4799-84d0-a78c295c4e2b
facade | 2025-06-16 14:51:31.758 [FCD] 172.18.0.1 - - [16/Jun/2025 14:51:31] "POST /messages HTTP/1.1" 200 -
facade | 2025-06-16 14:51:33.657 [FCD] Received POST: ImmutableMultiDict({'txt': 'message14'})
facade | 2025-06-16 14:51:33.657 [FCD] Requesting URLs from http://config-server:8883/services/logging
facade | 2025-06-16 14:51:33.662 [FCD] Retrieved URLs for Logging: ['http://logging1:50051/logging', 'http://logging2:50052/logging', 'http://logging3:50053/logging']
facade | 2025-06-16 14:51:33.662 [FCD] Attempting to send message id=a55544e1-e4c9-4c86-bb18-8cdf909cb995
facade | 2025-06-16 14:51:33.662 [FCD] Attempt 1 to http://logging1:50051/logging
facade | 2025-06-16 14:51:37.466 [FCD] Error sending to http://logging1:50051/logging: HTTPConnectionPool(host='logging1', port=50051): Max retries exceeded with url: /logging (Ca
used by NameResolutionError("curllib3.connection.HTTPConnection object at 0x7fca57bb0220: Failed to resolve 'logging1' ([Errno -2] Name or service not known)"))
facade | 2025-06-16 14:51:37.466 [FCD] Waiting 0.5s before retry
facade | 2025-06-16 14:51:37.967 [FCD] Attempt 2 to http://logging2:50052/logging
facade | 2025-06-16 14:51:41.759 [FCD] Error sending to http://logging2:50052/logging: HTTPConnectionPool(host='logging2', port=50052): Max retries exceeded with url: /logging (Ca
used by NameResolutionError("curllib3.connection.HTTPConnection object at 0x7fca57bb2260: Failed to resolve 'logging2' ([Errno -2] Name or service not known)"))
facade | 2025-06-16 14:51:41.759 [FCD] Waiting 1.0s before retry
facade | 2025-06-16 14:51:42.760 [FCD] Attempt 3 to http://logging3:50053/logging
facade | 2025-06-16 14:51:42.772 [FCD] Successfully sent to http://logging3:50053/logging
facade | 2025-06-16 14:51:42.772 [FCD] Stored message id=a55544e1-e4c9-4c86-bb18-8cdf909cb995
facade | 2025-06-16 14:51:42.773 [FCD] 172.18.0.1 - - [16/Jun/2025 14:51:42] "POST /messages HTTP/1.1" 200 -
PS D:\dist-sys\Task>
```